

[Λ] Arkham

# Data Science Challenge

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LMP Anomaly Detection

across Monclova Load Zone

**Santiago Velasco García**

Aspiring Forward Deployed Engineer Internship



## INTRODUCTION

# Santiago Velasco

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Hybrid Profile: Computer Engineering Student (UNAM)  
specialized in data.

**Key Experience (DiDi Food):** Building automated data workflows and operational frameworks. Focused on translating complex technical requirements into resilient, scalable solutions driven by business outcomes.

**About me:**

- Animal lover
- I play guitar and piano

# The Challenge & Executive Solution

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## The Operational Challenge

Analyze the volatility of Local Marginal Prices (LMP) across the 9 nodes in the Monclova Load Zone via the CENACE API.

The electricity market exhibits extreme asymmetry, causing "alert fatigue" if conventional static thresholds are used to measure risk.

## The Technical Deliverable

A production-ready ETL pipeline, orchestrated via CLI and built under software engineering standards.

Implements an **analytical consensus model** to classify critical events, automatically generating interactive reports.

# Resilient Architecture (ETL Pipeline)

☰ **Robust Extraction:** HTTP sessions with exponential backoff retries to handle API instability gracefully.

```
def extract_cenace_data(config: PipelineConfig, node_blocks: list[list[str]])  
    """Executes requests to the CENACE API and consolidates the data."""
```

🔑 **Src Layout & CLI:** Leveraging argparse and dataclasses for highly parameterizable execution.

```
def build_arg_parser() -> argparse.ArgumentParser:  
    parser = argparse.ArgumentParser(description="PML Anomaly Detection Pipeline (CENACE)")
```

🔒 **Reproducibility:** Dependency management with uv. The uv.lock file guarantees identical environments.

```
import argparse  
import logging  
import os  
import time  
from dataclasses import dataclass, field  
from datetime import date, datetime, timedelta  
from typing import Any  
  
import numpy as np  
import pandas as pd  
import plotly.express as px  
import plotly.graph_objects as go  
import requests as rq  
import warnings  
from requests.adapters import HTTPAdapter
```

☰ **Evaluation:** Automated calculation of the Jaccard Index to measure consensus without ground truth data.

```
def compute_evaluation_metrics(master_df: pd.DataFrame) -> pd.DataFrame:  
    """Calculates unsupervised evaluation metrics per node."""
```

# Architecture & Data Flow

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## CENACE API

SW-PML web service, 9  
Monclova nodes, SIN /  
MDA



## Extraction

Zone-to-node resolver,  
7-day chunks, retries with  
backoff



## Detection

Z-Score, Isolation Forest  
and IQR run per node



## Consensus

Critical Alert only if 2+  
methods agree

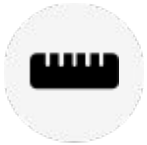


## Outputs

CSV datasets and  
interactive HTML reports

**Data Completeness Check:** *every response is validated node-by-node against the request — a status="OK" alone is not treated as proof that all requested nodes returned data.*

# Methodology: The "Analytical Trident"



## Robust IQR

Static Baseline. Interquartile range (x3.0) to identify historically extreme outliers across the time series.



## Rolling Z-Score

Dynamic Statistics. 168-hour rolling windows to detect immediate shocks, adapting to summer seasonality.



## Isolation Forest

Machine Learning. Unsupervised algorithm detecting multivariate inconsistencies by crossing price and hour.

**Business Rule:** An event is flagged as a *Critical Alert* only if there is consensus (2+ models agree).

# Key Results: 18 Months, 9 Nodes

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**118,152**

hours of LMP analyzed

9 nodes × 547 days × 24h

**2,035**

critical alerts flagged

consensus of 2+ methods

**0.08-0.18**

inter-method Jaccard agreement

low overlap → each method flags a distinct signal

IQR

**5.7%**

of hours flagged

Robust static baseline

Z-Score

**2.9%**

of hours flagged

Dynamic, seasonally adaptive

Isolation Forest

**1.0%**

of hours flagged

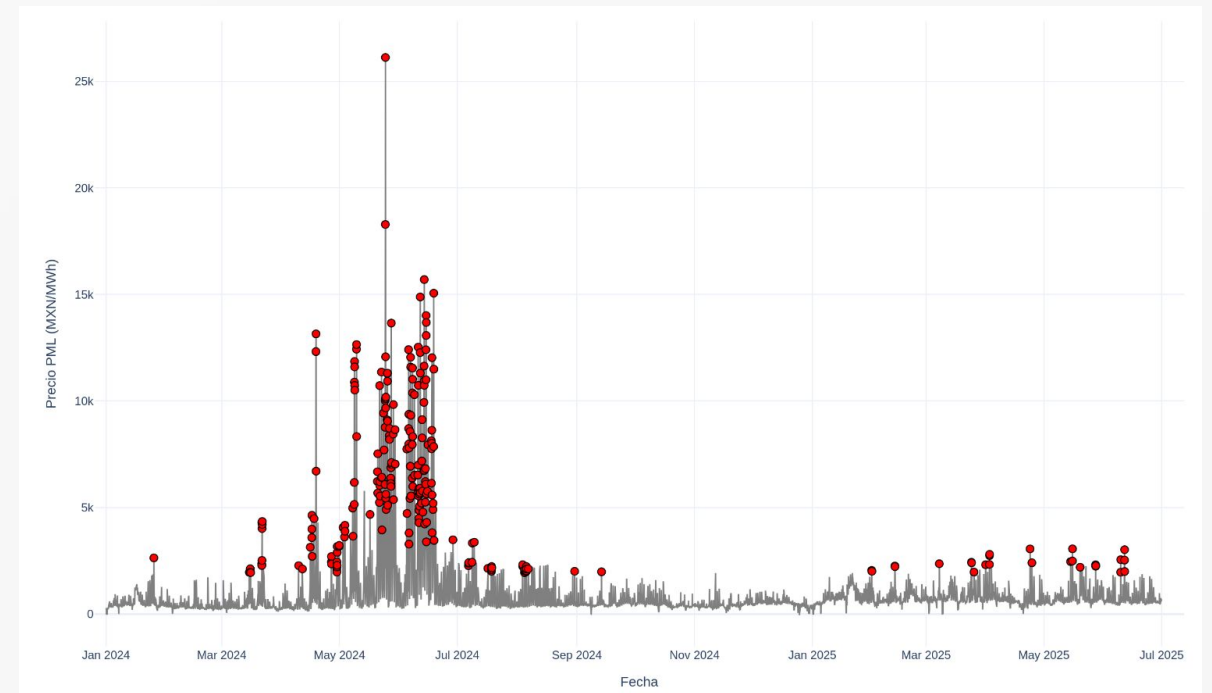
Multivariate ML detector

**Unsupervised Validation:** *no labeled ground truth exists for this market, so alerts were validated by internal consistency: flagged hours show a mean  $|Z\text{-score}|$  of 3.35 vs. 0.72 for non-flagged hours — confirming genuine statistical extremes, not method noise.*

# Reporting Suite

The pipeline auto-generates HTML artifacts (Plotly) to answer immediate business questions:

- **Lane Map (Scatter):** Identifies whether a spike was an isolated local failure or a systemic shock.
- **Risk Thermometer (Heatmap):** Monthly view for financial forecasting and seasonality tracking.
- **Interactive Dashboard:** Micro-breakdown to validate the anatomy of each critical alert.





# Visual Decision Making

Mapa Operativo Global: Alertas Críticas en la Red Monclova



**Systemic Shock Confirmed:** All 9 nodes flagged critical alerts simultaneously between April–June 2024 — consistent with a regional/systemic event (e.g. network congestion), not an isolated single-node failure.

# Technical Roadmap (Next Steps)

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-  **1. Cloud Deployment (Docker):** Containerize the script for scheduled execution (e.g., Apache Airflow), enabling daily automated updates.
-  **2. From Reactive to Predictive:** Integrate time series models (Prophet or XGBoost) to forecast alerts in the Day-Ahead Market (MDA).
-  **3. Real-time Alerts:** Connect the Critical Alerts output with messaging APIs (Slack, Microsoft Teams) to notify operations teams instantly.
-  **4. Analyst Self-Service:** Analysts consume the interactive dashboard and monthly heatmap directly — no code required — and can adjust node/date scope via the CLI without touching the pipeline internals.

# Image Sources

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[https://static.vecteezy.com/system/resources/thumbnails/083/027/680/small\\_2x/abstract-system-visualization-with-nodes-and-connections-digital-network-structure-with-binary-code-elements-in-monochrome-style-data-flow-analytical-mapping-and-technological-architecture-vector.jpg](https://static.vecteezy.com/system/resources/thumbnails/083/027/680/small_2x/abstract-system-visualization-with-nodes-and-connections-digital-network-structure-with-binary-code-elements-in-monochrome-style-data-flow-analytical-mapping-and-technological-architecture-vector.jpg)

Source: [www.vecteezy.com](https://www.vecteezy.com)



[https://img.magnific.com/premium-vector/modern-glassmorphism-infographic-elements-set-business-data-visualization-dashboard-ui\\_439195-251.jpg?semt=ais\\_test\\_b&w=740&q=80](https://img.magnific.com/premium-vector/modern-glassmorphism-infographic-elements-set-business-data-visualization-dashboard-ui_439195-251.jpg?semt=ais_test_b&w=740&q=80)

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